

MyPact — Engineering & Milestone Report

Academic Onboarding Backend, MongoDB Atlas & Platform Architecture

Official Status Report | Generated for Project Lead | September 2026

1. EXECUTIVE SUMMARY

MyPact is an academic discipline and study streak accountability platform engineered for university and college students. This report summarizes the newly built backend infrastructure, MongoDB Atlas cloud database synchronization, academic onboarding engine (Institution, Faculty, Level, Target CGPA, and Courses), and our technical roadmap.

2. COMPLETED MILESTONES (WHAT WE DID)

' Academic Onboarding Backend Engine (POST /api/auth/onboarding)

Built full academic data persistence for students completing the onboarding flow:

- Institution & University: Captures universities/polytechnics (UNILAG, UI, FUNAAB, FUTA, Harvard, etc.)
- Faculty & Department: Saves specific fields of study (Computer Science, Medicine, Law, Engineering)
- Academic Level: Tracks academic year (100L through 500L, Clinicals, Postgraduate)
- Target CGPA Goal: Stores desired academic standing (e.g. First Class 4.50 - 5.00 / 4.85 CGPA)
- Course Catalog Seeding: Automatically initializes chosen courses in the MongoDB "courses" collection
- Enforcement Tiers: Configures discipline intensity (Mild, Strict, or Zero Excuses Enforcement)
- Initial Pact & Partner: Seeds student first study pact and emergency accountability partner details

' Dedicated Backend Service on Port 5000 with Atlas Cloud Database

Deployed high-speed Express + TypeScript backend running strictly on port 5000 (<http://localhost:5000>), connected live to MongoDB Atlas cluster0.jub2d53.mongodb.net/mypact_db with 0 demo/mock data.

' Full Frontend-to-Backend Reactive Wizard (/onboarding)

Updated Next.js onboarding page with interactive saving animation, error fallback handlers, local storage user caching, and confetti celebration that transitions smoothly into /dashboard.

' Production Build Validation & GitHub Deployment

Tested Next.js 16 build with 0 TypeScript/ESLint errors across all 23 routes; pushed cleanly to GitHub repository (<https://github.com/CodeWithHife/My-Pact.git>) for continuous Vercel deployment.

' Light Mode Default & UI Polish

Configured clean, high-clarity Light Mode as default across all pages with seamless Dark Mode toggle.

3. DATA FLOW & ONBOARDING ARCHITECTURE

1. Student Wizard (/onboarding)

%% % Step 1: Institution (UNILAG, FUTA, UI...) + Faculty + Level (100L-500L) + Target CGPA (e.g. 4.85)

%% % Step 2: Course Selection (Chemistry, Math, CS, Anatomy, Engineering...)

%% % Step 3: Enforcement Tier Selection (Mild / Strict / Zero Excuses) + Partner Details

%% % Step 4: First Study Pact Configuration (Duration + Daily Schedule)

%

%%

2. API Layer (POST /api/auth/onboarding)

%% % Authenticates session token or email identifier

%% % Updates User document in MongoDB Atlas with academic metadata & sets isOnboarded: true

%% % Automatically seeds Course records in MongoDB "courses" collection

%% % Automatically creates the Initial Task in MongoDB "tasks" collection

%

%%

3. Student Dashboard (/dashboard)

%% % Loads live student profile, university badge, CGPA goal tracker, and course cards.

4. UPCOMING ROADMAP (WHAT WE STILL NEED TO DO)

- [] Phase 2: Interactive Study Pact Creator Engine

Build the interactive dashboard modal where students add course-specific study pacts (e.g., CHM 201 - 6 hrs/week, MAT 301 - 4 hrs/week), set proof verification requirements (note uploads / math solve), and select study stakes.
- [] Phase 3: Proof-of-Work Verification (Gemini Vision AI / OCR & Timer)

Implement live proof submissions: photo uploads of handwritten notes analyzed by Gemini Vision AI to verify authenticity, time-stamped barcode check-ins, and deep focus timers.
- [] Phase 4: Accountability Partner Dispatch & Referee System

Allow study partners to receive automated email/WhatsApp/SMS proof alerts to approve or contest a daily study session.
- [] Phase 5: Real-Time CGPA & Academic Habit Analytics

Connect the live Dashboard with real-time target CGPA progress bars, weekly study hour charts, streak milestones, and peer leaderboard rankings.
- [] Phase 6: Consequence & Deadline Enforcement Engine

Deploy background jobs to monitor daily study cutoff times and trigger accountability consequences (reputation deduction, emergency partner alerts, or stake forfeiture).

CURRENT SYSTEM STATUS: ONBOARDING & AUTH INFRASTRUCTURE FULLY OPERATIONAL

All changes pushed to GitHub main. Proceeding next to Phase 2: Interactive Study Pact Creator Engine.